

WORK EXPERIENCE

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| <b>AI/ML Computational Science Specialist, Accenture</b>                                                                                                                                                                                                                                                                                             | 09/2026 – Present |
| • <b>Privacy-Safe Analytics Agent:</b> Architected an end-to-end analytics platform with ingestion, PII/profanity redaction, hybrid (BM25 + Vector) Azure AI Search across 100K+ documents, multi-agent reasoning layer, shipped via Docker, GitHub Actions CI/CD with OpenTelemetry tracing; cutting retrieval latency ~75%, analyst effort 60–70%. |                   |
| <b>AI/ML Computational Science Sr Analyst, Accenture</b>                                                                                                                                                                                                                                                                                             | 09/2024 – 08/2026 |
| • <b>Enterprise Data Q&amp;A Agent:</b> Engineered a text-to-SQL agent chain (Microsoft Copilot Studio → AI Foundry) that translates natural-language queries into T-SQL executed directly against Lakehouse SQL endpoints, enabling analytics on 10M+ records in <30s.                                                                              |                   |
| • <b>GenAI Business Doc Generator:</b> Designed and delivered a multi-agent RAG system with LangGraph + LoRA fine-tuned LLM for business doc generation; with LangSmith tracing, ROUGE evals; cutting drafting time 60–70%                                                                                                                           |                   |
| <b>Software Developer, Expro</b>                                                                                                                                                                                                                                                                                                                     | 09/2022 – 09/2024 |
| • <b>Well Performance Analytics Platform:</b> Implemented a PyTorch LSTM-autoencoder pipeline for anomaly detection on well-sensor time-series data, improving risk detection by ~20% and reducing analysis time by ~40%.                                                                                                                            |                   |
| • <b>Oil &amp; Gas Insights Web Application:</b> Designed a PySpark KPI pipeline on Azure Databricks using PCA, K-means, and Random Forest to derive KPI features from equipment telemetry, reducing manual reporting by ~50%.                                                                                                                       |                   |
| <b>Data Scientist, Ganit Labs</b>                                                                                                                                                                                                                                                                                                                    | 03/2021 – 09/2022 |
| • <b>Equipment Failure Prediction Model:</b> Developed predictive maintenance models on GCP for oilfield equipment with ~85% recall and ~30% less downtime.                                                                                                                                                                                          |                   |
| <b>Biomedical Engineer, Portea, AOV, Care Group</b>                                                                                                                                                                                                                                                                                                  | 08/2018 – 09/2020 |
| • Led cross-functional teams in medical equipment maintenance and training programs.                                                                                                                                                                                                                                                                 |                   |

EDUCATION

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| <b>MTech (Master of Technology), University of Calcutta</b> | 2016 – 2018 |
| <b>BTech (Bachelor of Technology), WBUT</b>                 | 2011 – 2015 |

TECHNICAL SKILLS

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| <b>Languages:</b><br>Python, SQL                                                                                                                                                   | <b>Generative AI &amp; Frameworks:</b><br>Multi-agent orchestration, Retrieval-Augmented Generation (RAG), Prompt engineering, LLM fine-tuning, Observability, Evaluation, LangChain, LangGraph, CrewAI, Claude Agent SDK, MCP |
| <b>AI Architecture &amp; Governance:</b><br>Solution architecture, multi-agent & RAG design patterns, AI guardrails, human-in-the-loop governance, cost/latency trade-off analysis | <b>Machine Learning &amp; Deep Learning:</b><br>LSTM autoencoders, Isolation Forest, PCA, K-means, HDBSCAN, Random Forest, Decision Tree, Linear Regression, Logistic Regression, NLP                                          |
| <b>Cloud Platforms:</b><br>Azure (AI Foundry, Storage, Fabric, Cosmos DB, AI Search, OpenAI, App Service, Function App, Logic Apps); GCP (Cloud Storage, BigQuery, App Engine)     | <b>Data &amp; Retrieval:</b><br>Vector search: FAISS, Pinecone, Azure AI Search · Graph: Neo4j · Platforms: Microsoft Fabric, Azure SQL, Azure Databricks                                                                      |
| <b>MLOps &amp; DevOps:</b><br>Docker, Git, GitHub Actions CI/CD, MLOps & LLMops pipelines, LangSmith, OpenTelemetry, OpenEvals, RAGAS, REST APIs                                   | <b>Frameworks &amp; Libraries:</b><br>PyTorch, scikit-learn, Hugging Face Transformers, Azure OpenAI / OpenAI SDK, FastAPI, Flask                                                                                              |

CERTIFICATIONS & TRAININGS

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| • <a href="#">Claude Certified Architect – Foundations</a> | • <a href="#">AI-102: Microsoft Azure AI Engineer Associate</a>  | • <a href="#">CS50AI: Introduction to Artificial Intelligence with Python — Harvard University</a> |
| • <a href="#">AZ-900: Microsoft Azure Fundamentals</a>     | • <a href="#">LLMOps, MLOps &amp; CI/CD/CT Pipelines (Udemy)</a> |                                                                                                    |

PROJECTS & PUBLICATIONS

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| <b><a href="#">rti-engine – AI Pay Transparency Compliance Engine</a></b> , <a href="https://github.com/trayan4/rti-engine">https://github.com/trayan4/rti-engine</a><br>Architected an 11-node LangGraph multi-agent system for pay-transparency compliance — 6 model-driven, 5 deliberately deterministic where no judgment is required — with tiered autonomy, human approval gates, cross-vendor review, figure-validation. MCP-exposed Neo4j graph; deployed on Azure, with Docker/GitHub Actions CI-CD, LangSmith/OTel tracing, openevals harness. |
| <b><a href="#">Autonomous Data Analyst</a></b> , <a href="https://github.com/trayan4/autonomous-data-analyst">https://github.com/trayan4/autonomous-data-analyst</a><br>Engineered a LangGraph multi-agent text-to-SQL system on Azure with governance-first routing, execution-repair validation, forced RCA decomposition, model tiering, and OpenTelemetry cost tracing; 87% on a 110-question golden set.                                                                                                                                            |
| <b><a href="#">Multimodal Agentic Reasoning Assistant (MARA)</a></b> , <a href="https://github.com/trayan4/MARA-AI-Agent">https://github.com/trayan4/MARA-AI-Agent</a><br>Designed a multi-agent system coordinating RAG, Vision, Data & Web Search agents (3–5 parallel agent calls per query); hybrid FAISS+BM25 retrieval (recall@10 ~0.75), adaptive planning, FastAPI (p95 ~30s).                                                                                                                                                                   |
| <b><a href="#">GPT-Style Language Model from Scratch</a></b> , <a href="https://github.com/trayan4/llm-from-scratch">https://github.com/trayan4/llm-from-scratch</a><br>Implemented a decoder-only 124M-parameter LLM from scratch in PyTorch — BPE tokenization, multi-head causal self-attention, training loop, instruction fine-tuning; trained on 124M tokens.                                                                                                                                                                                      |

AWARDS

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| <b>Spotlight Award, Client</b> — delivery of the privacy-safe analytics agent platform | 02/2026 |
| <b>Growth Catalyst Award, Accenture</b> — technical proficiency and project delivery   | 12/2025 |